3. Number of spindle speeds, number of feeds.
4. Spindle nose taper.
5. Power required
6. Nett weight of the machine
7. The floor space required
8. Type of the machine

5.7 Fundamental milling process

The various milling process may be grouped under two headings

1. Peripheral milling
2. Face milling

Peripheral milling

The machining is performed by the cutting edges on the periphery of the milling cutter. It is classified under two headings.

1. Up milling
2. Down milling.

Up milling

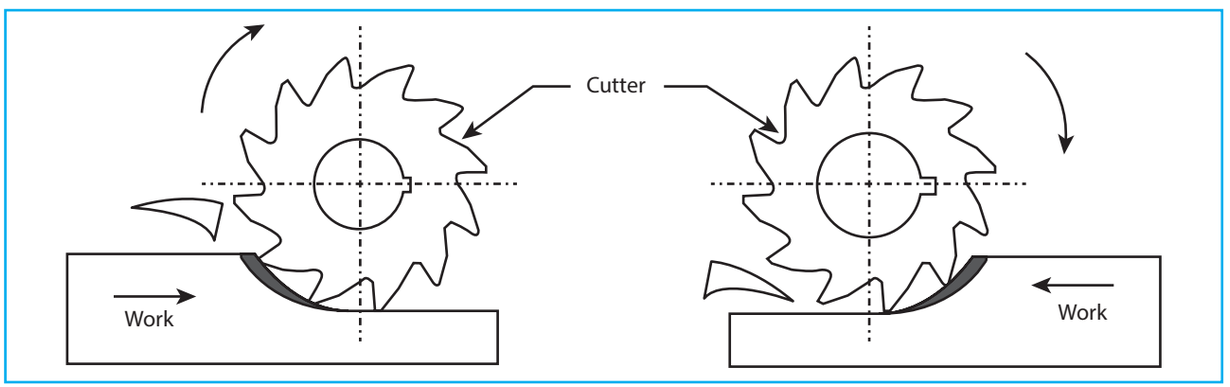
In this method, the workpiece mounted on the table is fed against the direction of rotation of the milling cutter. The cutting force is minimum during the beginning of the cut and maximum at the end of cut. The thickness of chip is more at the end of the cut. At the cutting force is directed upwards. It tends to lift the workpiece from the fixtures. A difficulty is felt in pouring coolant on the cutting edge. Due to these reason the quality of the surface obtained by this methods is wavy. This processes being safer is commonly used and sometimes called conventional milling.

Down milling

The workpiece mounted on the table is moved in the same directions as that of the rotation of the milling cutter. The cutting force is maximum at the beginning and minimum at the end of cut. The chip thickness is more at the beginning of the cut the workpiece is not disturbed because of the bite of the cutter on the work. The coolant directly reaches to the cutting point. So the quality of surface finish obtained is high. Because of the backlash error between the feed screw of the table and the nut, vibration is set up on the workpiece.

Face milling

During face milling, the machining is performed by the peripheral cutting edges. The surface obtained by the processes is perpendicular to the axis of rotation of the cutter.



Up Milling Down Milling

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